



Probability of Default Prediction for Banca Massiccia

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Business Understanding

Identify, Define, and Motivate

- Improve Banca Massiccia's loan underwriting process by developing accurate models to estimate the one-year probability of default for prospective borrowers.
- More accurate PD estimates can help the bank set risk-based pricing including appropriate interest rates and underwriting fees while minimizing the risk of loss.
- Existing models may lack the sophistication to identify nuanced risk factors, leading to missed or overestimated defaults.

Data Mining Solution

By mining historical loan data and financial statements, the solution extracts patterns and relationships between borrower characteristics and default risk.

- Financial metrics are derived to represent borrower financial health.
- Leverages rare events (Default) using machine learning models like XGBoost, ensuring the predictive power of the data is maximized.
- Uses clustering and calibration to refine predictions in sparse default datasets.

This allows the bank to better estimate the likelihood of default for prospective borrowers, optimizing capital reserves.



Problem Formulation

Finance Perspective

- Historically, banks have relied on traditional credit scoring models like logistic regression and discriminant analysis for credit scoring. Examples including Altman's Z-score and Moody's RiskCalc.
- These models, while interpretable, often struggle with non-linear relationships and complex interactions between variables.

Machine Learning Perspective

- Neural Networks (eg., Tam and Kiang (1992)) showed improved accuracy over traditional models but lacked interpretability.
- Logistic Regression and Penalized Variants (eg., Dumitrescu et al. (2022)) showed small performance gain and also lacked interpretability.
- Random Forests (eg., Breiman, L. (2001)) offered better performance with imbalanced datasets.



Improvements in Our Approach

Data Utilization:

- Data Sparsity: inspired by the Shumway model (2001), use data that spans over several years in order to obtain a suitable sample for modeling.
- Data Quality: Implements data cleaning and imputation techniques, including cluster-based and finance-driven imputation methods, to effectively handle missing values and ensure robust model performance.

Interpretability:

- Integrate financially motivated metrics such as debt-to-equity ratio, operating cash flow ratio, and gross profit margin, ensuring alignment with credit risk theory and regulatory frameworks.
- By using SHAP values and feature importance analysis, we make machine learning models interpretable, providing actionable insights.

Data Understanding

Data Overview

- The given dataset is in size of **1023552 entries** and **44 columns**.
- To address our problem, we selected **13 independent variables** from original dataset for later features engineering and produce **1 new target variable** to apply the supervised learning algorithms later:
 - Independent Variables = ['eqty_tot', 'asst_tot', 'liab_lt', 'liab_lt_emp', 'roa', 'rev_operating', 'COGS', 'cf_operations', 'AP_lt', 'debt_lt', 'debt_fin_lt', 'debt_bank_lt', 'asst_current']
 - Target Variable = ['default']

What is “Default”?

- We define the unit of analysis in our project as a firm year, which means that each row in our dataset representing one firm year.
- Based on our previous reading, a firm year typically starts in April and lasts for 12 months.
- However, since our dataset only provides the statement date (December 31st) for each record of data, we adjusted the firm year-end by adding 12 months plus an additional 3 months to the statement date. This adjustment ensures that the firm year aligns with the April start and March end in practice.

Data Understanding

What is “Default”? (Cont.)

- To create the target variable, **default**, we determined whether a default event (def_date) occurred within the duration of the firm year. Specifically, we check that for each row:
 - If the **def_date** is null, it indicates that no default events occurred. As a result, we assigned a value of 0 to the default variable.
 - If the **def_date** is not null and occurred before the firm year-end, we assigned a value of 1 to the default variable. This indicates that a default event happened during the firm year.
 - If the **def_date** is not null but occurred after the firm year-end, we assigned a value of 0 to the default variable because the default falls outside the given firm year time frame.
- After applying this approach, the number of default (1) and non-default (0) events are shown as below:
 - Number of rows with **default=1: 14114**
 - Number of rows with **default=0: 1009438**

Data Understanding

Derived Features

- The features used in our modeling are all financial ratios derived from the 13 selected variables that mentioned earlier.
- These features are carefully chosen to represent the following 4 key financial dimensions:
 - **Solvency:** Reflects the percentage of company financing that comes from creditors and investors. Higher ratios are usually viewed as more risky.
 - $\text{Equity Ratio} = \text{Total Liabilities} / \text{Total Equity}$
 - $\text{Total Liabilities} = \text{Total Assets} - \text{Total Equity}$
 - **Profitability:** Reflects how effectively a company uses its assets to generate profit and compares product pricing to its basic cost. Two features are used here because we think profitability would take large impact in predicting probability of default.
 - ROA
 - $\text{Gross Profit Margin} = (\text{Operating Revenue} - \text{COGS}) / \text{Operating Revenue}$

Data Understanding

Derived Features (Cont.)

- **Liquidity:** Determine how much cash a business has on hand at any given time that it can use to pay off its liabilities.
 - Operating Cash Flow Ratio = Operating Cash Flow / Current Liabilities
 - $\text{Current Liabilities} = \text{Total Liabilities} - (\text{Long-term Liabilities} + \text{Long-term Liabilities to Employees} + \text{Long-term Bank Debt} + \text{Long-term Debt Other} + \text{Long-term Accounts Payable} + \text{Long-term Debt Other})$
 - An alternative ratio that we did not use in modeling is Current Ratio = Current Asset / Current Liabilities. Not use due to high collinearity with other features.
- **Size:** The scale of a company's assets in a given firm year
 - Total Assets

Data Understanding

Problems

- **Missing Values:** After investigating the 13 selected variables, we find out that most of them are containing large number of missing values as shown below:

eqty_tot	1
asst_tot	0
liab_lt	294
liab_lt_emp	162
roa	27
rev_operating	174
COGS	24
cf_operations	88
AP_lt	92244
debt_lt	155
debt_fin_lt	92068
debt_bank_lt	90971
asst_current	1
dtype:	int64

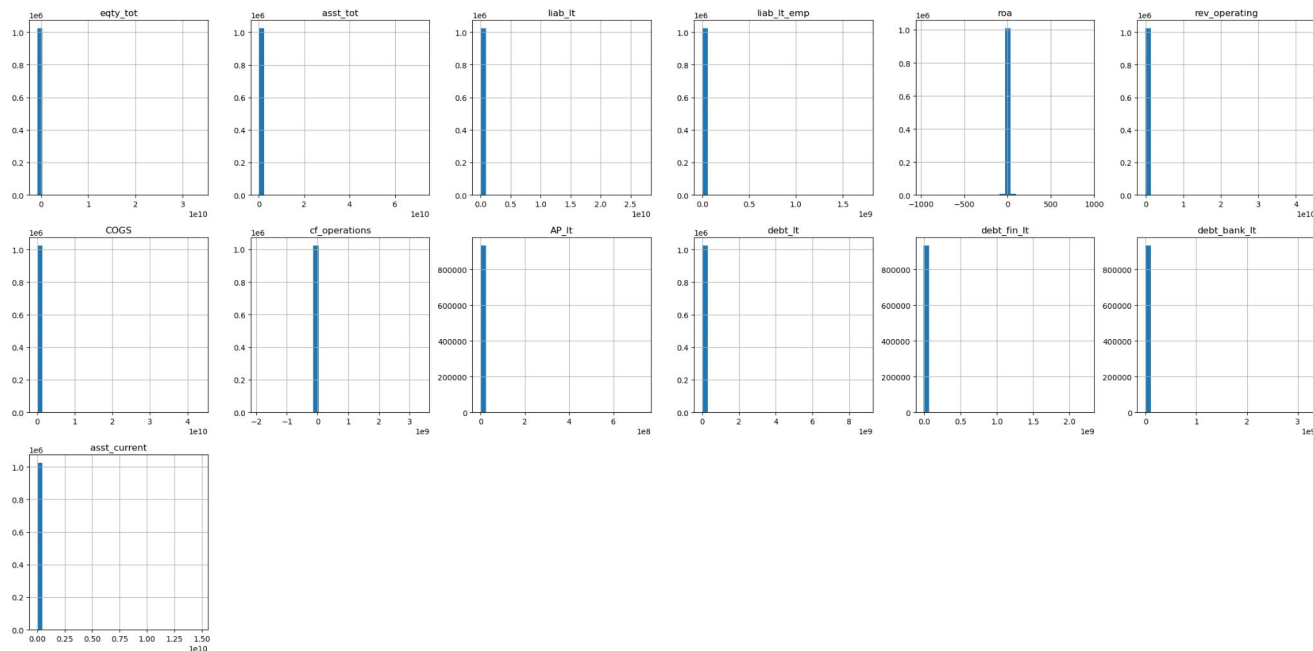
- **Denominator Variables Equal to 0:** When calculating financial ratios in the later step, zero values in the denominator variables will resulted in problematic value as inf or -inf.

eqty_tot	15
asst_tot	0
liab_lt	641396
liab_lt_emp	331579
roa	2277
rev_operating	31217
COGS	647
cf_operations	1073
AP_lt	910254
debt_lt	281542
debt_fin_lt	851880
debt_bank_lt	451100
asst_current	170
dtype:	int64

Data Understanding

Problems (Cont.)

- **Highly Skewed Data:** Many of the selected variables exhibited highly skewed distributions as shown below:



Data Understanding

Limitation & Potential Bias

- **Limitation of Variables:** Since the dataset only includes variables from financial statements, we are unable to use additional financial dimensions that require variables outside of the current dataset. For example, the efficiency dimension that reflects the effectiveness of firm's management of its resources will need variables like inventory, which is not included in the current dataset. Due to this lack of variables, the estimation may be less comprehensive compare to a real-world setting.
- **Time Bias:** The given dataset contains only data from a specific time period. This means that the results estimated from this dataset might not be able to generalize to different times periods or different economic environments.
- **Data Selection Bias:** The bank may only have historical data from companies whose loan requests have been approved. This means that the dataset is likely excludes information from companies whose applications have been rejected, which may highly correlated to the probability of default prediction. This create a potential data bias.

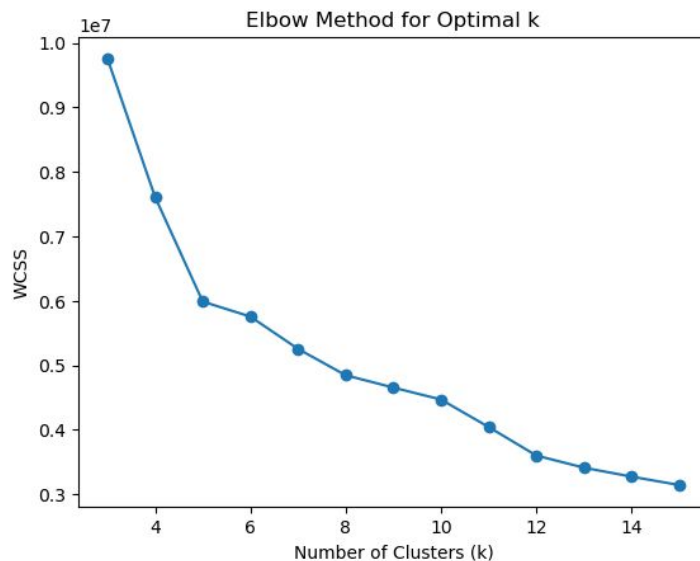
Data Preparation

Datetime Conversion

We converted all time related variables into the correct format “datetime64”.

Handling Missing Values

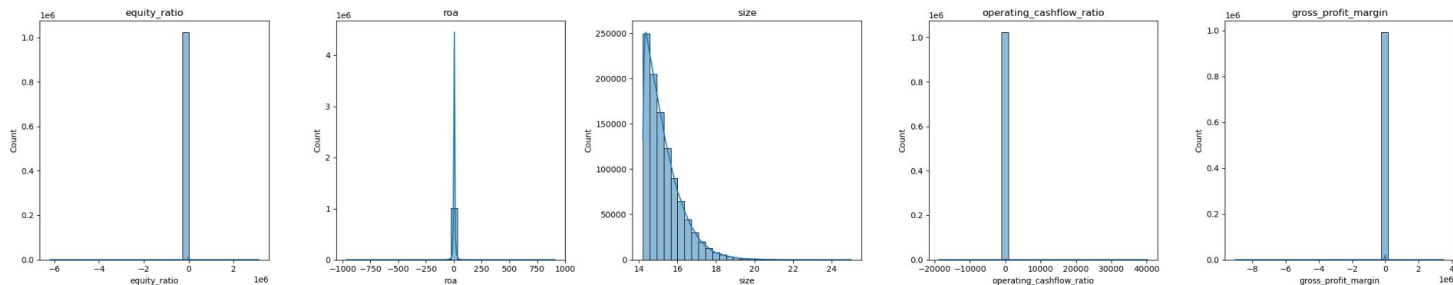
To handle missing values in the selected variables dataset, we used a clustering-based imputation method. After first standardizing the data using StandardScaler, we applied the K-Means clustering algorithm to identify groups of similar data points and determine the optimal number of clusters (5) by using the Elbow Method. We then replaced the missing values in each variable with the median of that variable within the corresponding cluster. This approach makes sure that the imputed values are able to reflect the overall patterns within similar data groups



Data Preparation

Log Transformation

- We only did log transformation on feature “**size**” (also the variable “**asst_tot**”) to avoid scaling problem in the later modeling step.
- We did not apply log transformation to any original selected variables to handle the skewness problem because these variables would be used to further calculate the financial ratios in subsequent steps. Applying log transformation would change the meaning of the resulting financial ratios in the real world setting.
 - A simple example here will be $\frac{2}{3} \neq \log(2) / \log(3)$.
- However, we later found out that calculating the financial ratios will somehow handle the skewness of these variables.



Data Preparation

Addressing Infinite Values (inf or -inf)

- To address infinite values (inf and -inf) in the resulting dataset after calculating the financial ratios, we replaced them with the median of the corresponding feature.
- By using this median imputation methods, the inf or -inf values will not influence the overall distribution of the features and preserve the central tendency without introducing bias or outliers.

Collinearity Check with VIF

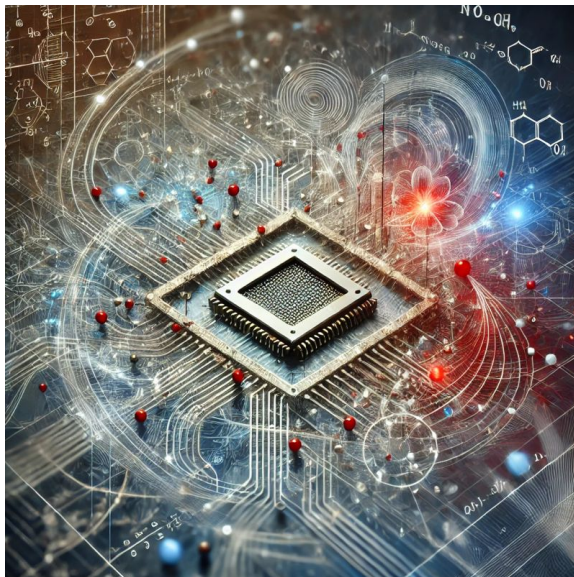
To ensure the features contribute to the model without redundancy, we also calculate the Variance Inflation Factor (VIF) for each feature. The results are shown as below and suggest that there is no significant multicollinearity among the features in the dataset:

	feature	VIF
0	equity_ratio	1.000006
1	roa	1.080945
2	size	1.080496
3	operating_cashflow_ratio	1.000046
4	gross_profit_margin	1.000773

Data Preparation

Limitation

- **Computational Complexity:** The clustering-based imputation method can be computationally intensive for large dataset. Although it works in our current dataset, it might not be available for dataset with higher dimensionality.



Modeling

Selected Models

- **Decision Tree Classifier:** Selected as a baseline for its interpretability and ability to highlight feature importance.
 - Pros: simple; interpretable; quick implementation
 - Cons: prone to overfitting; limited scalability for complex relationships
- **XGBoost Classifier:** Chosen for its superior performance with large datasets and robustness in handling missing values.
 - Pros: high accuracy; ability to handle missing values; flexibility in parameter tuning
 - Cons: computationally intensive; requires careful tuning to avoid overfitting

Alternative Models

- **Linear Regression:**
 - Linear assumptions, which might oversimplify relationships in financial data
 - Sensitive to missing values
- **Random Forest:**
 - lacked the efficiency and flexibility of gradient boosting

Modeling

Final Set of Variables Used

- **Profitability:** Return on Assets (ROA), Gross Profit Margin
- **Solvency:** Debt-to-Equity Ratio
- **Liquidity:** Operating Cash Flow Ratio
- **Size:** Total Assets

Economic Intuition of the Model

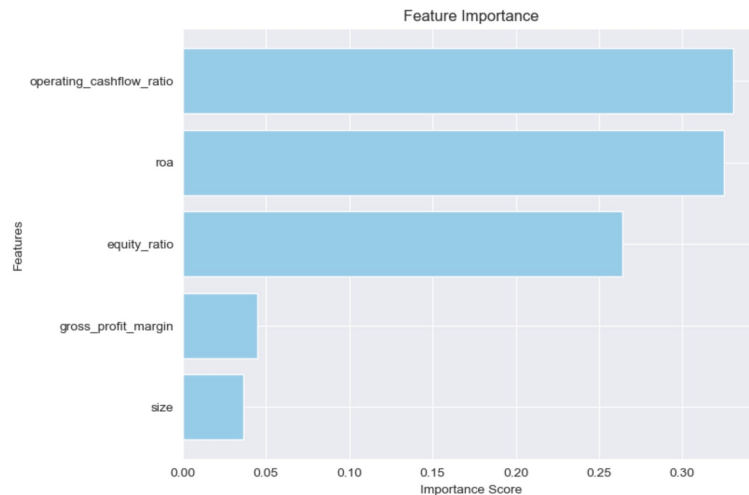
The selected variables align with economic intuition, as they represent:

- A firm's ability to generate profit (profitability metrics).
- Financial health and leverage (solvency indicators).
- Ability to meet obligations with cash flow (Liquidity ratios).

These features are critical for predicting defaults or financial distress, enabling targeted risk management.

Feature Importance:

	Feature	Importance
3	operating_cashflow_ratio	0.330323
1	roa	0.324786
0	equity_ratio	0.263903
4	gross_profit_margin	0.044621
2	size	0.036366



Modeling

Final Model Specification

XGBoost Classification Model

Hyperparameters:

- Learning rate (learning_rate): 0.1
- Maximum tree depth (max_depth): 5
- Number of estimators (n_estimators): 100
- Subsample ratio (subsample): 0.8
- Colsample_bytree: 1.0

Performance:

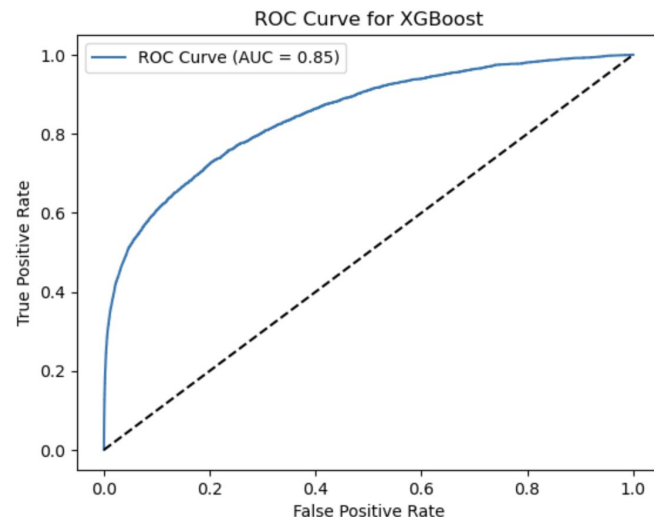
- ROC-AUC: 0.85
- Confusion Matrix:

	Predicted Non-Default	Predicted Default
Non-Default	302480	352
Default	3669	565

Model Performance

The ROC-AUC score of 0.85 indicates the model's strong ability to classify risks, distinguishing between positive (default) and negative (non-default) cases with high accuracy.

By leveraging this performance, the firm can confidently assess financial default risk, ensuring efficient allocation of resources.



Evaluation

Objective and Key steps:

Evaluate the performance of the model and compare it with benchmarks through

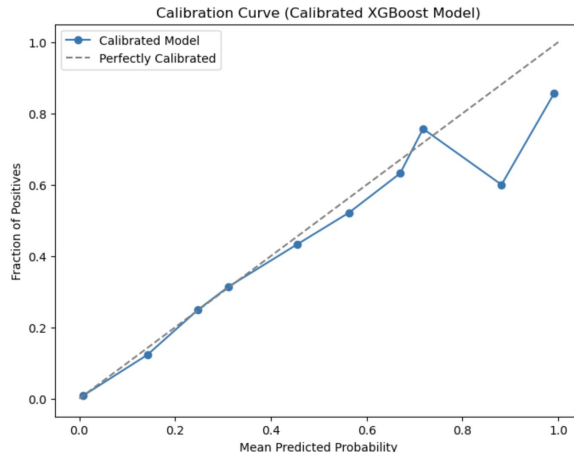
- Cross-validation with CalibratedClassifierCV.
- Calibration curve analysis to assess probability predictions.
- Benchmarking against uncalibrated models and baseline model.
- Metrics: ROC-AUC

Model	ROC-AUC
Decision Tree	0.81
XGBoost	0.85

Benchmarks and Calibration Curve

We use decision tree models as a benchmark for comparing XGBoost models. Decision trees are a common choice for baseline models

- They are simple, easy to understand and implement.
- They provide a transparent way to see how input features are split at different levels to make decisions, which can be very insightful, especially in business environments where interpretability is important.



- This blue line plots the actual positivity rate for each partition against the average predicted probability for that partition. The closer this line is to the dotted line, the better the model is at predicting the true probability.
- The model appears to be fairly well calibrated, but there are some deviations between the predicted probabilities of 0.8 - 1. This might be due to imbalanced, sparse data of default.

Evaluation

Business Case Development

To project expected improvement, we should consider tangible and measurable outcomes, focusing on financial and operational metrics.

Expected Improvement:

- Reduced default rates (Cost savings from reduced defaults and better pricing)
- Improved risk-based pricing strategies (Projected improvement in loan underwriting profitability)
- Portfolio performance (possibly through calculation of Expected Loss through PD estimation, more data needed)

Challenges and Alternatives

Challenges:

- Limited historical data for rare events (defaults) and for historical ROI for similar machine-learning implementations in the bank.
- Difficulties in quantifying ROI due to external economic factors. Loan performance is influenced by macroeconomic conditions, such as interest rates and market downturns, which are outside the model's control.
- Implementation costs might be high due to training and managing fee.

Alternatives:

- Automation improves underwriting speed.
- Conduct scenario analysis and stress testing for additional insights.
- Use SHAP values for interpretability and regulatory alignment.

Deployment Process

- **Integration into Loan Underwriting System:** The model has to work seamlessly with the existing financial and risk management systems. The model's predictions will be integrated into the bank's existing loan underwriting workflow, providing loan officers with a PD for each prospective borrower.
- **Automated decision-making for low-risk borrowers:** loans with low PD can be automatically approved, reducing processing time and operational costs.
- **Risk monitoring and portfolio management:** post-approval, the model can be used to monitor changes in borrowers' financial conditions and update PD estimates. Perform stress testing regularly can ensure model resilience.
- **Testing in a controlled environment:** test model in a sandbox environment using historical data with known outcomes to verify that the model's predictions align with actual default events and that there are no system integration issues. Pilot the model in phases, proceeding to full deployment after performance and impact are observed.
- **User training and documentation:** training sessions for loan officers and comprehensive documentation on how the model works, its limitations, and how to interpret the output.
- **Monitor and Maintain:** The performance of the model is regularly monitored to allow any degradation over time. The model is also reviewed regularly and updated with new data and due to changes in financial conditions.

Ethical or Legal Considerations

Fairness and Bias: Ensure the model doesn't discriminate against any group or entity. This will also include identification and mitigation of any biases within the data or model processing. Legal compliance, particularly with regulations like the Equal Credit Opportunity Act (ECOA) in the U.S

Data Privacy: The data protection laws should be followed, like GDPR in Europe or CCPA in California, while using and keeping the sensitive financial data.

Transparency: Some level of transparency on the part of the model in how it makes its predictions, especially where those are potentially consequential for the companies being evaluated.

Settings Unsuitable For Model Application

Highly volatile markets (Economic Instabilities):

- Highly volatile markets, such as those that experience rapid price swings as a result of economic crises, geopolitical events or major policy changes. During the 2008 financial crisis, for example, many forecasting models failed because they could not predict the size of the recession and its impact on company performance.

Emerging Industries companies:

- Emerging industries or industries undergoing significant innovation, such as renewable energy or tech start-ups. These industries lack historical data and have untested business models, making forecasts less reliable.

Companies Undergoing Restructuring:

- A listed company in an industry undergoing a major restructuring may exhibit unbalanced financial results in terms of profit, operating profit and financial profit. This may occur as a result of restructuring charges, changes in operational focus or asset revaluation

Companies in Seasonal Businesses:

- If some of the companies in the dataset are in seasonal industries, metrics such as rev_operations and cogs may show significant fluctuations that are not representative of annual performance, which can cause damage to model predictions.

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- Course Website, Cheat Sheets, Finance-related

Contribution of Members

Analysis:

- Data preprocessing
 - EDA: Chenxing Li, Lia Wang, Maggie Xu
 - Missing Value Imputation: Chenxing Li, Lia Wang
 - Feature Engineering: Maggie Xu, Lia Wang, Haiyi Yue, Chenxing Li
- Modeling and Evaluation: Haiyi Yue
- Calibration: Lia Wang, Maggie Xu
- Harness Function: Maggie Xu, Lia Wang

Write-up:

- Business Understanding: Maggie Xu
- Problem Formulation: Maggie Xu
- Data Understanding: Lia Wang
- Data preparation: Lia Wang
- Modeling: Haiyi Yue
- Evaluation: Chenxing Li, Maggie Xu
- Deployment: Chenxing Li